MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR

EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCOM301 - COST ACCOUNTING

Duration 2 Hours (2024 ADMISSION ONWARDS) Maximum Marks:70

PART A

Very Short Answers (Answer any 12 Questions Each carries 1 marks)

1. A cost centre is a location, person, item of equipment, department or other unit for which costs are separately collected and measured.

2. Indirect labour.

3. Under-absorption of overheads.

4. Cost Accounting Standards (CAS).

5. Cost accountancy.

6. Other employee benefits / allowances and benefits.

7. Decreases, other things remaining constant.

8. Normal idle time.

9. Selling. (CAS 15 is the Cost Accounting Standard on Selling and Distribution Overheads.)

10. Normal loss.

11. Arrival and Departure of Workers /Attendance of workers

12. Factory lighting – factory overhead; office salary – administration overhead; advertisement – selling overhead; carriage outward – distribution overhead.

13. No purchase consideration / cost incurred by the receiving entity; any valuation should follow the applicable costing policy and circumstances.

14. Group bonus system / group incentive scheme.

PART B

Short Answer questions answer any 4 questions each question carries 7 Marks

15. Major methods of costing and their applicability

Costing methods are selected according to the nature of production, the product or service, and how costs can be identified. The major methods are:

Method	Meaning / application	Typical example
Unit / output costing	Cost per unit is determined where output is homogeneous and continuous.	Cement, bricks, sugar, paper
Job costing	Costs are accumulated separately for each specific job or customer	Printing, repair workshops, custom furniture

	order.	
Batch costing	A batch is treated as a cost unit; batch cost is divided by units in the batch.	Pharmaceuticals, garments, components
Contract costing	Costs are collected for each large, long-term contract, often at a site.	Buildings, roads, bridges, shipbuilding
Process costing	Costs are accumulated process-wise for continuous, repetitive production; average cost per process/unit is found.	Chemicals, oil refining, textiles
Operation costing	Costs are accumulated for a defined operation or series of operations, often for standardised components.	Engineering and component manufacture
Operating / service costing	Cost per service unit is calculated for service organisations.	Transport per passenger-km, hospitals per patient-day, hotels per room-day
Multiple / composite costing	Two or more methods are combined when a product has many components or stages.	Cars, machinery, complex equipment

16. Gantt Task and Bonus Scheme

Note: Value liberally, full marks may be awarded for genuine attempt in any method of wage calculation The Question not separately state the Time Wage Rate

17. Machine Hour Rate

Computation of machine hour rate

	Per annum	Per Hour
A. Standing Charges: 1.Rent (2,000 × 12 months × 1/5) 2.Insurance 3.Supervisor Salary(2,500 × 12 months × 1/5)	 4800 1000 6000	
Standing Charges per hour (11800/2000)	11800	5.9
B. Machine Expenses: 1.Depreciation ((1,20,000 + ₹10,000 − ₹5,000) ÷ 10) =12500/2000 2.Repairs (2000/2000) 3.Power (6*1)		 6.25 1.00 6.00
Machine Hour rate		**19.15**

*** Annual working hours. =2000**

18. Economic Order Quantity (EOQ) and frequency of orders (Value Liberally)

Annual demand (A) = 60,000 units;
Ordering cost per order (O) = 120;
Unit price = 40;
Return on investment = 12%;
Other carrying charges = 8 per unit per annum.

Annual carrying cost per unit (C) = 12% of 40 + 8 = 4.80 + 8 = 12.80.

EOQ = √ (2AO/C).
EOQ = √ [(2 × 60,000 × 120) ÷ 12.80] = √1,125,000 = 1,060.66 units, (**1,061 units**.)
Number of orders per year = 60,000 ÷ 1,060.66 = **56.57 orders.**
Frequency in days (using a 365-day year) = 365 ÷ 56.57 = 6.45 days between orders.

19. Labour turnover and its significance in cost accounting

Labour turnover is the rate at which employees leave an organisation and are replaced during a specified period. It indicates the extent of labour movement in and out of the organisation.

Common measurement methods:

• **Separation method** = (Number of separations during the period ÷ Average number of workers) × 100.

• **Replacement method** = (Number of replacements during the period ÷ Avg. number of workers) × 100.

• **Flux method** = [(Separations + Replacements) ÷ Average number of workers] × 100.

Average number of workers is commonly calculated as (Opening workers + Closing workers) ÷ 2, unless monthly averages are available.

Significance: **(1)** high turnover increases recruitment, selection and training costs; **(2)** inexperienced workers may cause lower output and more defects; **(3)** production schedules can be disrupted; **(4)** overtime and replacement costs may rise; **(5)** it helps management assess labour stability, morale and working conditions;

A moderate turnover due to normal retirement or movement may be unavoidable; the reasons should be analysed rather than interpreting the percentage in isolation.

20. Compute works cost

Given: direct materials ₹10,000; direct wages ₹30,000; direct expenses ₹10,000. Indirect wages = 50% of direct wages; indirect materials = 50% of direct materials; indirect expenses = 100% of direct expenses.

Computation of Works Cost

Particulars		**Amount (₹)**
Direct materials	10000	
Direct wages	30000	
Direct expenses	10000	
Prime Cost		**50000**
Add: Factory overheads:		
Indirect materials (50% × 10,000)	5000	
Indirect wages (50% × 30,000)	15000	
Indirect expenses (100% × 10,000)	10000	**30000**
Works / Factory Cost		**80000**

PART C

Answer any 2 Questions Each Question Carries 15 marks

21. Apportionment of service department overheads by simultaneous equation method

Overhead cost asper primary Distribution Summary:

P1 = 800, P2 = 700, P3 = 500, S1 = 234, S2 = 300.

The allocation percentages in the paper are interpreted as follows:

	P1	**P2**	**P3**	**S1**	S2
S1	20%	40%	30%		10%
S2	40%	20%	20%	20%	

(1) Calculation of total overheads of Service departments

Let total cost of Service dept... S1 be **X** and total cost of S2 be **Y**

X = 234 + 20% of y = 234 + 0.20y

Y = 300 + 10% of x = 300 + 0.10x

Substitute y into the first equation: X= 234 + 0.20(300 + 0.10x) = 234 + 60 + 0.02x.

Therefore 0.98x = 294, and X = 300. Then Y = 300 + 0.10(300) = 330.

Overheads of S1 dept. =X= 300

Overheads of S2 dept. =Y= 330

(2)Overheads Chargeable to Production departments

Over heads of S1 dept. = 300

Less overhead Chargeable to S2 Dept. = 30 (300*10/100)

=300-30**=270**

Over heads of S2 dept. = 330

Less overhead Chargeable to S2 Dept. = 66 (330*20/100)

= 330-66**=264**

(3) Statement showing secondary Distribution

Particulars	**Ratio**	**P1**	**P2**	**P3**
Overheads as per Primary Distribution	------	800	700	500
Share from S1 (270)	**(2:4:3)**	60	120	90
Share from S2 (264)	**(4:2:2)**	132	66	66
Total Overheads		992	886	656

22. Cost statement of XYZ Ltd. for June 2026

Elements of Cost		Total
Direct materials: Opening Stock Of materials Add: Purchase of materials	 70000 500000	
	570000	
Less: Closing Stock of materials	(55000)	
(1)Material Consumed		**515000**
Direct wages		**260000**
(2) Prime Cost		**775000**
Factory Overheads: Add: opening stock of Work in Progress Less: Closing stock of Work In Progress		110000 15000 (20000)
(3)Factory / Works Cost		**880000**
Administrative Overheads		**60000**
(4) Cost Of Production		**940000**
Add: Opening finished goods Less: Closing finished goods		95000 (120000)
(5) Cost of goods sold		**915000**
Add: Selling and distribution overheads		30000
(6)Cost of sales		**945000**
(7)Net profit (BF)		**155000**
Sales		**1100000**

23. Wages under Halsey 50%, Halsey–Weir and Rowan incentive plans

Standard time = 12 hours;

Actual time = 9 hours;

Wage rate = ₹80 per hour.

Time saved = 12 − 9 = 3 hours.

Time wages for actual time = 9 × ₹80 = ₹720.

Standard time wages = 12 × ₹80 = ₹960.

(a) Halsey Bonus Plan (50%)

Wages= (Time Taken * Rate) + 50 %(Time Saved * Rate)

= (9*80) +50 %(3*80)

Total earnings = 720+120 = **₹840.**

(b) Halsey–Weir Premium Plan

The question does not state the premium percentage. The standard textbook convention of $33\ ^{1/3}$ % is assumed

Wages= (Time Taken * Rate) + $33\ ^{1/3}$ %(Time Saved * Rate)

= (9*80) + $33\ ^{1/3}$ % (3*80)

Total earnings= 720+(240*1/3)= **₹800**

(c) Rowan Plan

Wages = **(Time Taken * Rate) + [** $\frac{Timetaken*Rate*TimeSaved}{(Standardtime)}$]

(9*80)+ [$\frac{9*80*3}{(12)}$] = 720+180 = 900

Total earnings = ₹720 + ₹180 = **₹900**.

Comparison Table

Plan	Actual time wages (₹)	Bonus (₹)	Total earnings (₹)
Halsey (50%)	720	120	840
Halsey–Weir ($33\ ^{1/3}$% assumed)	720	80 (72)	800 (792)
Rowan	720	180	900

Rowan gives the highest earnings here (**₹900**), followed by Halsey 50% (**₹840**), then Halsey–Weir at 30% (**₹800**). The worker saves 3 hours under all three plans. Under Rowan, the bonus is linked to the proportion of time saved relative to standard time; under Halsey and Halsey–Weir, it is a fixed percentage of the wages for time saved.

***Halsey –Weir premium Plan :** the bonus rate ranges from **30% to 33** $^{1/3}$ **%** of the wages of the time saved

Award full marks for correct calculation using either **30% or 33** $^{1/3}$ **%**

Wages= (Time Taken * Rate) + 30 %(Time Saved * Rate)

= (9*80) + 30 % (3*80)

Total earnings= 720+72= **₹792**